RTRP-A-07

PROJECT APPROVAL

WHEELS FOR WOMEN

Real Time Research Project Abstract Approval submitted in partial fullfilment of the requirements for the award of the degree

BACHELOR OF TECHNOLOGY In INFORMATION TECHNOLOGY

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DEPARTMENT OF INFORMATION TECHNOLOGY

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Problem Statement:

Despite the growing popularity of riding among women, there remains a lack of dedicated platforms that address the unique needs and challenges faced by female riders. These challenges include limited access to female-centric riding communities, insufficient safety resources, and a lack of tailored information about riding gear and techniques. Additionally, female riders often encounter feelings of isolation or insecurity, especially in male-dominated riding spaces. This gap in the market creates a need for a platform that not only connects women with similar interests but also provides resources for safety, skill development, and community engagement. The Female Riding app aims to address these issues by offering a comprehensive, user-friendly platform designed specifically for women riders.

Objective:

- **To provide safe and reliable transportation services for women**: Ensuring that women have access to trustworthy and secure transportation options.
- **To promote women's mobility and independence:** Empowering women to move freely and participate fully in economic, social, and cultural activities.
- **To address transportation-related challenges faced by women:** Identifying and addressing specific transportation needs and concerns of women, such as safety, affordability, and accessibility.
- **To create a supportive and inclusive transportation community for women:** Fostering a sense of community and connection among women who use the transportation network.
- **To promote women's economic empowerment through transportation**: Providing women with access to transportation that can help them access employment, education, and other economic opportunities.

Existing System:

- Public Transportation: Buses, trains, and subways that operate on fixed routes and schedules.
- Taxi Services: Traditional taxi companies that provide door-to-door service.
- Ride-hailing Services: Companies like Uber and Lyft that offer on-demand transportation.
- Private Vehicles:Women may use their own vehicles or those of family and friends

Disadvantages:

- Safety Concerns: Women may feel uncomfortable or unsafe using public transportation or ride hailing services, especially at night.
- Limited Accessibility: Public transportation may not be accessible or convenient for women with disabilities or those living in areas with limited transportation options.
- Cost:Taxi services and ride-hailing services can be expensive, especially for women who need to travel frequently.
- Lack of Reliability: Public transportation and ride-hailing services may not always be reliable, leading to delays or cancellations.

Proposed System:

The proposed system for the Female Riding app is designed to create a comprehensive and supportive platform for women riders. It would feature personalized user profiles, allowing riders to connect with like-minded individuals, and a community space for group discussions and event organization. Safety features, such as real-time navigation, emergency alerts, and location tracking, would ensure secure riding experiences. The app would also offer educational resources on riding techniques, gear, and safety, as well as a marketplace for women's riding gear. Overall, the system aims to provide a space where female riders can connect, learn, and ride with confidence.

Advantages:

- **Safety and security**: Female riding services provide a safe and secure environment for women to travel, especially at night or in unfamiliar areas.
- **Convenience**: Female riding services offer a convenient mode of transportation for women, especially those who may not have access to personal vehicles or prefer not to drive themselves.
- **Empowerment**: Female riding services promote women's empowerment by providing them with independence and mobility, enabling them to travel freely and participate in various activities.
- **Reduced harassment**: Female riding services reduce the risk of harassment and unwanted attention that women may face while traveling alone.
- **Increased mobility**: Female riding services increase women's mobility, enabling them to travel to work, school, or other destinations without relying on male family members or friends.

- **Job opportunities**: Female riding services create job opportunities for women, enabling them to work as drivers, dispatchers, or other support staff.
- **Social benefits**: Female riding services provide social benefits, such as enabling women to connect with others, build relationships, and participate in social activities.
- **Health benefits**: Female riding services promote physical activity, such as cycling or walking, which can improve women's health and well-being.
- **Environmental benefits**: Female riding services can promote sustainable transportation options, such as cycling or electric vehicles, which can reduce carbon emissions and promote environmental sustainability.
- **Increased confidence**: Female riding services can increase women's confidence and self-esteem, enabling them to feel more independent and empowered

Modules:

- **User Registration and Profile Management**:

  Allow users to create accounts, upload profile pictures, and manage personal details.

  Include verification processes to ensure user authenticity and safety.

- **Ride Booking and Scheduling**:

  Enable users to book immediate rides or schedule future rides.

  Provide options for ride preferences (e.g., solo rides, women drivers only).

- **Safety Features**:

  Implement real-time location tracking for rides.

  Add an SOS button that alerts emergency contacts and local authorities in case of danger.

  Include a feature for users to share their ride details with friends or family.

- **Driver Management**:

  Allow drivers to register, complete training, and get verified.

  Provide tools for drivers to view and manage ride requests.

- **Payment Integration**:

  Support multiple payment options such as credit/debit cards, digital wallets, and in-app payments.

Include features for payment history and generating ride receipts.

- **Ratings and Reviews**:

  Allow users and drivers to rate and review each other to maintain quality service.

  Implement a feedback system to address issues and improve user experience.

Functional Requirements:

User Management

- Registration
- Profile Management
- Login/Logout

Ride Booking

- Ride Request
- Ride Matching
- Ride Confirmation

Payment Management

- Payment Gateway
- Fare Calculation
- Payment Processing

Safety features

- Emergency Button
- Real-time Tracking
- Driver Verification

Rating and Review

- User Rating
- Driver Rating

Admin Panel

- User Management
- Driver Management
- Ride Management
- Payment Management
- Reporting and Analytics

Non-Functional Requirements:

These are basically the quality constraints that the system must satisfy according to the project contract. The priority or extent to which these factors are implemented varies from one project to other. They are also called non-behavioral requirements. They basically deal with issues like:

- Performance
- Scalability
- Reliability
- Security
- Availability
- Maintainability
- Compatibility
- Accessibility
- Responsiveness

Hardware Requirements:

- Processor: Quad-core processor (e.g., Intel Xeon or AMD Ryzen)
- Memory: 16 GB RAM (or more)
- Storage: 1 TB hard drive (or more)
- Operating System: 64-bit operating system (e.g., Windows Server, Linux, or macOS)

Software Requirements:

- IDE : Visual studio code
- Server : Apache Tomcat
- Database server :MySQL
- UML Design ER MOdelling Tools:Star UML
- Front-end : HTML, CSS,Javascript
- Back-end : Python, Java Script nodejs
- Mapping services : Google Maps
- Payment Gateway : Gpay, PhonePe, Paytm, cash